

TidyTuesday Week 32

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Tuesday 4th August, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the segment where a computer flings two unrelated economic time series into a blender and we all pretend to be surprised by the smoothie. This week was a light one – only two matchups made it to the arena – but honestly, both of them earned their spot in the highlight reel, one for overachieving and one for face-planting so spectacularly that I felt genuine secondhand embarrassment.

First up, August 3rd gave us the crossover event nobody demanded: Coinbase Bitcoin versus the San Francisco Tech Pulse, an index so thoroughly retired it's been discontinued since 2020. The raw regression came out swinging with an R-squared of 0.70 and a p-value so small it looks like a typo – roughly 2 times ten to the negative seventeenth, which is math's way of screaming. "Bitcoin explains 70 percent of Bay Area tech vibes!" you might yell, before I gently remind you that both of these things went up over time, and going-up correlates with going-up. Truly the discovery of the century. But here's the mildly interesting twist: once we detrended both series and stripped out their shared upward drift, the relationship didn't fully die. It sagged to an R-squared of about 0.14 with a p-value around 0.003 – still technically "significant." So this is that rarer beast, a correlation that survives the trend-ectomy. Does that mean Bitcoin and San Francisco's tech mood share some deep cosmic bond? Almost certainly not. But it clung to life, and in this project that's basically a standing ovation.

Then came August 4th to restore the natural order of the universe. We paired Philadelphia's "Current Physical Plant Capital Expenditures Diffusion Index" – a title so long it needs its own zip code, popularity rating: a heartbreaking 1 – against the "Real User Cost Index of MSI-M2M," which stopped reporting in 2013 and has a popularity of 4, making this a genuine battle of the wallflowers. The raw regression produced an R-squared of 0.0006 and a p-value of 0.89, which is statistics politely saying "absolutely nothing is happening here." The detrended version somehow did even worse, with a p-value of 0.94. Two obscure indices, walking past each other in the dark, waving at completely different people. Beautiful. Chef's kiss. A perfect null result.

As always, the mandatory disclaimer that I am legally and morally obligated to shout from the rooftops: these are randomly paired series with zero causal story and no rigorous claim attached. Spurious correlation isn't the risk here – it's the entire business model. So no, do not rebalance your portfolio based on a discontinued San Francisco tech gauge, and please do not tell your Philadelphia capital-expenditure friends that the M2M cost index has forsaken them. It's all noise wearing a nice suit. See you next week for more.

2 Date: 2026-08-03

Series ID: CBBTCUSD

This series is titled Coinbase Bitcoin and has a frequency of Daily, 7-Day. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2014-12-01 and the observation end date is 2026-08-02. The popularity of this series is 75.

Series ID: SFTPINDM114SFRBSF

This series is titled San Francisco Tech Pulse (DISCONTINUED) and has a frequency of Monthly. The units are Index Jan 2000=100 and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1971-04-01 and the observation end date is 2020-03-01. The popularity of this series is 10.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_SFTPINDM114SFRBSF	R-squared:	0.695
Model:	OLS	Adj. R-squared:	0.690
Method:	Least Squares	F-statistic:	139.1
Date:	Mon, 03 Aug 2026	Prob (F-statistic):	2.21e-17
Time:	14:09:51	Log-Likelihood:	-185.92
No. Observations:	63	AIC:	375.8
Df Residuals:	61	BIC:	380.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	84.9060	0.863	98.348	0.000	83.180	86.632
value_fred_CBBTCUSD	0.0018	0.000	11.796	0.000	0.002	0.002

Omnibus:	9.975	Durbin-Watson:	0.305
Prob(Omnibus):	0.007	Jarque-Bera (JB):	10.825
Skew:	0.715	Prob(JB):	0.00446
Kurtosis:	4.442	Cond. No.	8.11e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 8.11e+03. This might indicate that there are strong multicollinearity or other numerical problems.

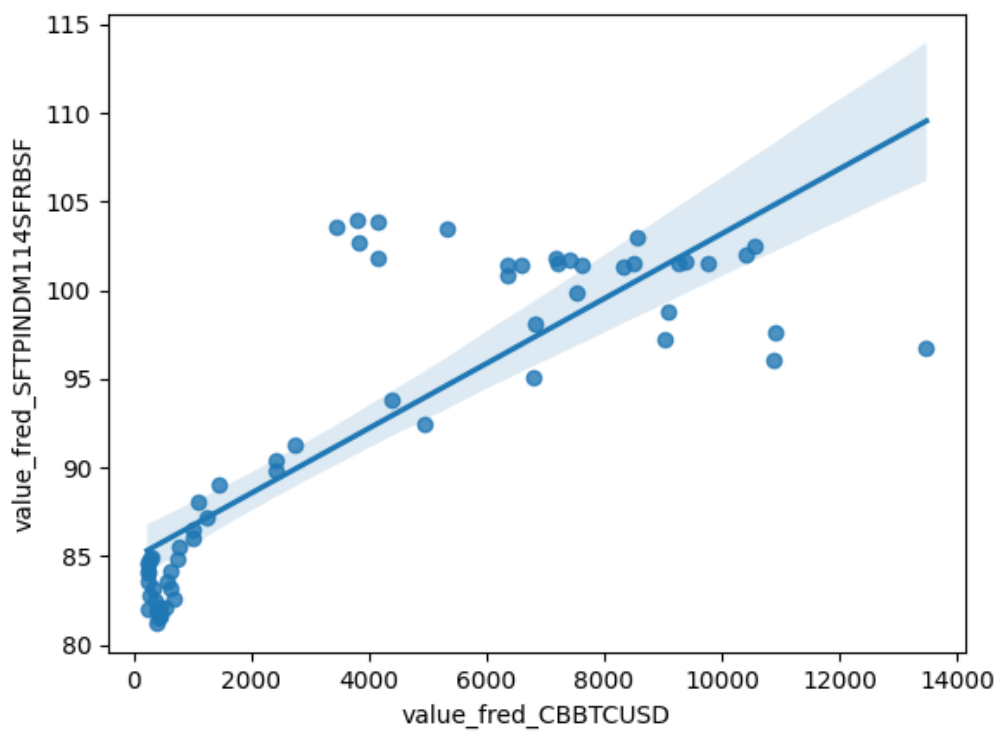


Figure 1: Raw Regression Plot for 2026-08-03

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_SFTPINDM114SFRBSF_detrended	R-squared:	0.137
Model:	OLS	Adj. R-squared:	0.123
Method:	Least Squares	F-statistic:	9.672
Date:	Mon, 03 Aug 2026	Prob (F-statistic):	0.00284
Time:	14:09:51	Log-Likelihood:	-150.74
No. Observations:	63	AIC:	305.5
Df Residuals:	61	BIC:	309.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	3.857e-12	0.339	1.14e-11	1.000	-0.678	0.678
value_fred_CBBTCUSD_detrended	0.0005	0.000	3.110	0.003	0.000	0.001

Omnibus:	23.048	Durbin-Watson:	0.095
Prob(Omnibus):	0.000	Jarque-Bera (JB):	4.918
Skew:	0.258	Prob(JB):	0.0855
Kurtosis:	1.732	Cond. No.	2.25e+03

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 2.25e+03. This might indicate that there are strong multicollinearity or other numerical problems.

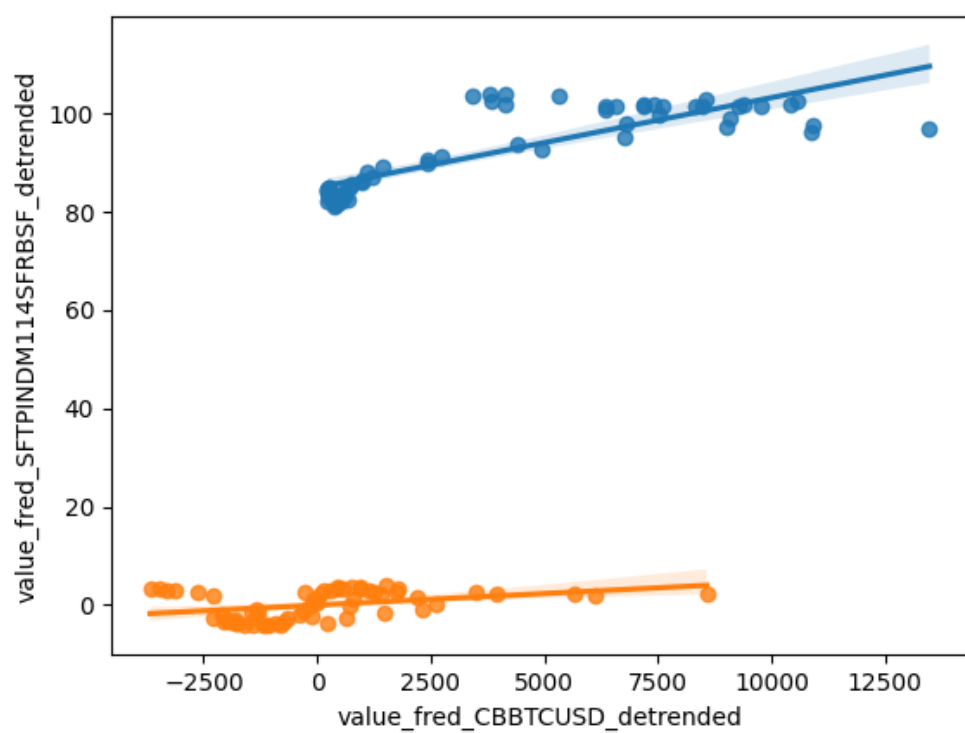


Figure 2: Detrended Regression Plot for 2026-08-03

3 Date: 2026-08-04

Series ID: CPBNDIF066MSFRBPHI

This series is titled Current Physical Plant Capital Expenditures; Diffusion Index for Federal Reserve District 3: Philadelphia and has a frequency of Monthly. The units are Index and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2011-03-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

Series ID: OCM2MP

This series is titled Real User Cost Index of MSI-M2M (preferred) and has a frequency of Monthly. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1967-01-01 and the observation end date is 2013-12-01. The popularity of this series is 4.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_OCM2MP	R-squared:	0.001
Model:	OLS	Adj. R-squared:	-0.031
Method:	Least Squares	F-statistic:	0.01870
Date:	Tue, 04 Aug 2026	Prob (F-statistic):	0.892
Time:	13:31:41	Log-Likelihood:	184.86
No. Observations:	34	AIC:	-365.7
Df Residuals:	32	BIC:	-362.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	0.0155	0.000	36.337	0.000	0.015	0.016
value_fred_CPBNDIF066MSFRBPHI	3.385e-06	2.47e-05	0.137	0.892	-4.7e-05	5.38e-05

Omnibus:	9.983	Durbin-Watson:	0.076
Prob(Omnibus):	0.007	Jarque-Bera (JB):	2.548
Skew:	0.142	Prob(JB):	0.280
Kurtosis:	1.689	Cond. No.	39.7

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

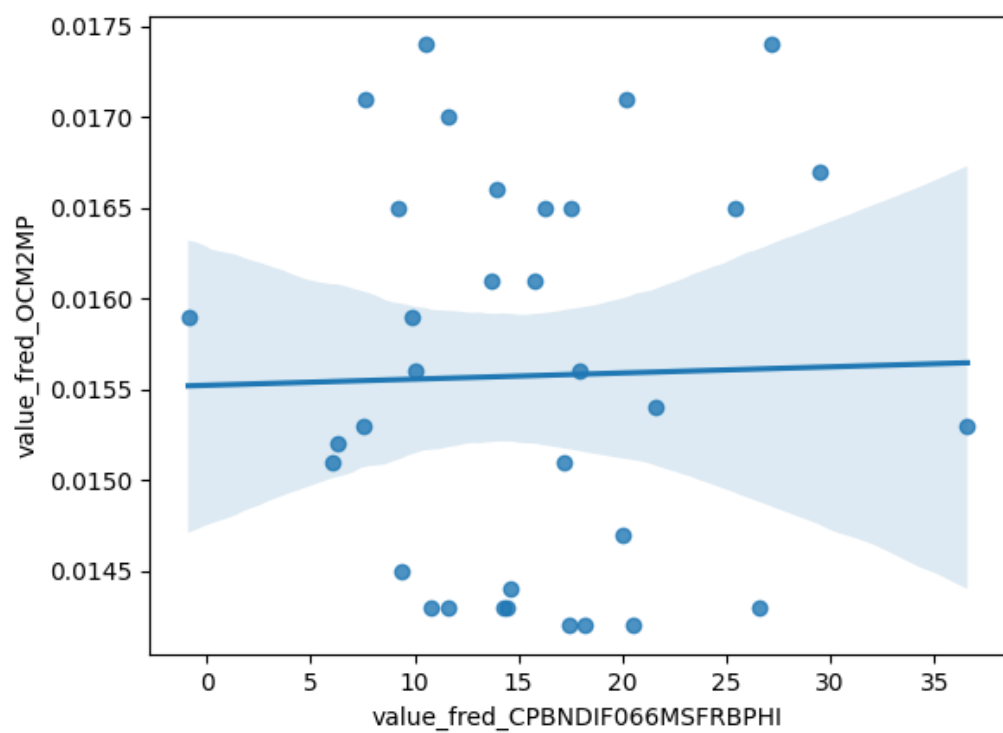


Figure 3: Raw Regression Plot for 2026-08-04

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_OCM2MP_detrended	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.031
Method:	Least Squares	F-statistic:	0.006047
Date:	Tue, 04 Aug 2026	Prob (F-statistic):	0.939
Time:	13:31:41	Log-Likelihood:	193.17
No. Observations:	34	AIC:	-382.3
Df Residuals:	32	BIC:	-379.3
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.858e-15	0.000	3.33e-11	1.000	-0.000	0.000
value_fred_CPBNDIF066MSFRBPHI_detrended	1.508e-06	1.94e-05	0.078	0.939	-3.8e-05	4.1e-05

Omnibus:	10.150	Durbin-Watson:	0.122
Prob(Omnibus):	0.006	Jarque-Bera (JB):	2.467
Skew:	0.016	Prob(JB):	0.291
Kurtosis:	1.681	Cond. No.	7.52

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

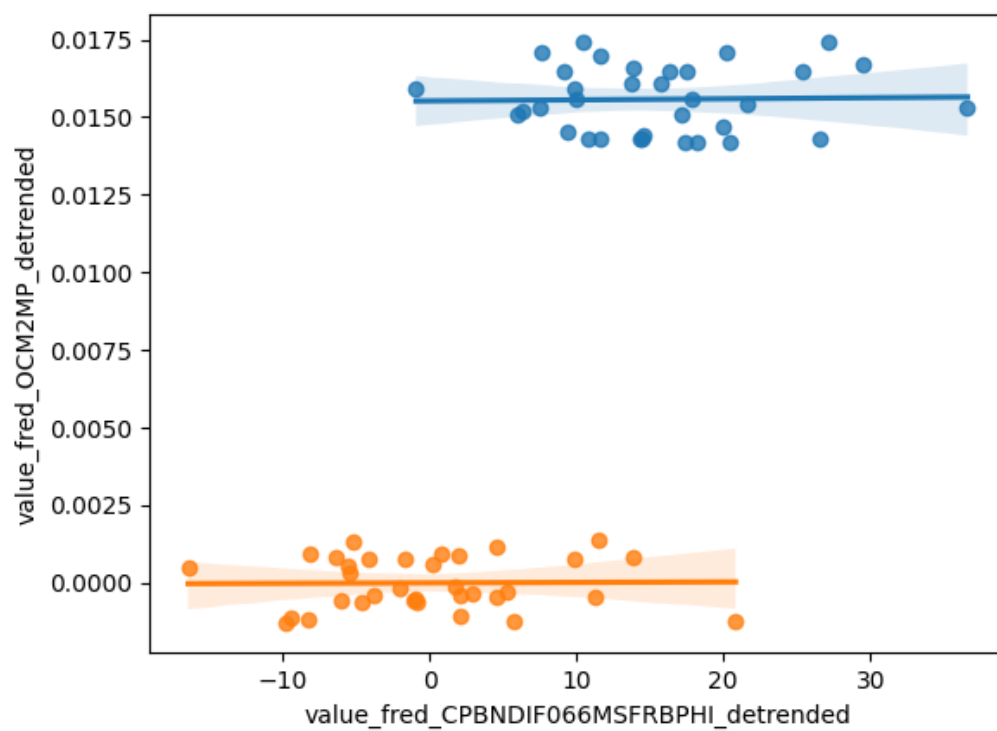


Figure 4: Detrended Regression Plot for 2026-08-04